

DOLFINx Thermo-Mechanical Simulation

A Comprehensive Learning Guide

Quick Reference & Decision Making

1. THE PROBLEM: What Are We Simulating?

Physical Setup:

A stainless steel plate ($1\text{m} \times 0.15\text{m} \times 0.02\text{m}$) is clamped on the left edge. A candle applies $20,000\text{ W/m}^2$ of heat to the center of the left edge for 600 seconds. We want to know:

1. How hot does the plate get?
2. Where is it hottest?
3. How much does it expand/deform?

2. GEOMETRY & MESH

Parameter	Value	Meaning
Length (L)	1.0 m	How long the plate is
Width (W)	0.15 m	How wide the plate is
Thickness	0.02 m	How thick the plate is (2 cm)
nx (horizontal cells)	100	Divides the plate into 100 pieces horizontally
ny (vertical cells)	20	Divides the plate into 20 pieces vertically
Total triangles	4,000	$100 \times 20 \times 2 =$ number of finite elements

Why mesh? The computer divides the plate into tiny triangles. It solves the physics equations on each triangle, then assembles the results.

3. MATERIAL PROPERTIES (304 Stainless Steel)

Property	Symbol	Value	Units	What It Means
Density	ρ	8000	kg/m^3	How heavy the material is
Thermal Conductivity	k	16.2	$\text{W}/(\text{m}\cdot\text{K})$	How fast heat spreads (higher = faster)
Specific Heat	c_p	500	$\text{J}/(\text{kg}\cdot\text{K})$	Energy needed to raise temp 1°C
Young's Modulus	E	193×10^9	Pa	How stiff/rigid the material is
Poisson Ratio	ν	0.29	—	How much it squeezes sideways when stretched
Thermal Expansion	α	17.3×10^{-6}	$1/\text{K}$	How much it expands per $^\circ\text{C}$

4. CANDLE PROPERTIES & POSITION

Property	Value	Meaning
Heat Flux	$20,000\text{ W/m}^2$	Power output per square meter (VERY hot!)
Candle Height	20 mm	The flame is 20mm tall
Position	Left edge, centered	Centered vertically on the left edge
Duration	600 seconds	Burn for 10 minutes

Key: The candle applies a NEUMANN boundary condition (specified FLUX, not temperature)

5. BOUNDARY CONDITIONS - THE DECISION TREE

What's a Boundary Condition?

A rule that describes what happens at the edges of your domain. It tells the solver: 'At this boundary, you must satisfy this condition.'

Two Main Types:

Type	Definition	You Control	Example	In This Code
DIRICHLET	Specify the VALUE	The value itself	Temperature = 100°C	Left edge position (u=0)
NEUMANN	Specify the FLUX/RATE	The flow or force	Heat flux = 20,000 W/m ²	Candle heat, convection

Decision Tree: Which Condition to Use?

START: Look at a boundary

- ↓
- **Is it CLAMPED/FIXED/HELD?** YES → DIRICHLET (u = 0) NO → Continue...
- ↓
- **Is there a FORCE/FLUX APPLIED?** YES → NEUMANN (specify F or q) NO → Continue...
- ↓
- **Is it EXPOSED to SOMETHING?** YES → Check what:
 - Ice water at 0°C → DIRICHLET (T=0)
 - Air at 24°C → ROBIN/NEUMANN (q=h·ΔT)NO → Continue...
- ↓
- **Is it FREE/UNCONSTRAINED?** YES → NEUMANN (natural: force=0)

6. IN YOUR CODE - THERMAL BOUNDARIES

Boundary 1: Candle Region (ds(5)) - LEFT EDGE, CENTERED

Aspect	Value
Boundary	Left edge, 20mm centered
Marker	ds(5)
Type	NEUMANN (flux specified)
Equation	L_thermal += q_candle × thickness × v_T × ds(5)
Physics	Heat input from candle
Why Neumann?	You CONTROL the heat flux (20,000 W/m ²), not the temp
Code	L_thermal += q_candle * thickness * v_T * ds(5)

Boundary 2: Convection Edges (ds(1,2,3,4)) - ALL OTHER EDGES

Aspect	Value
Boundaries	Left (no candle), Right, Top, Bottom
Markers	ds(1), ds(2), ds(3), ds(4)
Type	ROBIN/NEUMANN (heat transfer law)
Equation	a_thermal += h × T × v_T × thickness × ds(1,2,3,4) L_thermal += h × T_ambient × v_T × ds(1,2,3,4)
Physics	Cool to ambient (24°C) by convection

Why Neumann?	You CONTROL ambient temp & convection coefficient
The Law	Heat flux = $h \times (T_{\text{surface}} - T_{\text{ambient}})$
h value	10 W/(m ² ·K) - weak convection to still air

Boundary 3: Top & Bottom Surfaces - ENTIRE SURFACES

⚠ **Special consideration:** The plate has TWO surfaces (top and bottom). Each loses heat by convection.

Aspect	Value
Integration	Over VOLUME (dx) not just boundary (ds)
Type	ROBIN/NEUMANN
Factor	2.0 (because 2 surfaces)
Equation	$a_{\text{thermal}} += 2.0 \times h \times T_{\text{trial}} \times v_T \times dx$ $L_{\text{thermal}} += 2.0 \times h \times T_{\text{ambient}} \times v_T \times dx$
Physics	Cool through thickness on top AND bottom

Why use dx instead of ds? Because you're integrating over the THICKNESS, not just the edge.

7. IN YOUR CODE - MECHANICAL BOUNDARIES

Left Edge: CLAMPED (DIRICHLET)

Aspect	Value
Boundary	Left edge
Type	DIRICHLET (value specified)
Condition	$u_x = 0, u_y = 0$ (can't move)
Code	<code>bc = fem.dirichletbc(u_zero, left_dofs, V_u)</code>
Physics	Plate is clamped in a vise
Why Dirichlet?	You CONTROL the position (fixed)

Right Edge & All Other Edges: FREE (NEUMANN - IMPLICIT)

Aspect	Value
Boundaries	Right, top, bottom
Type	NEUMANN (implicit)
Condition	No boundary condition applied = naturally free
Physics	Can move freely based on internal stresses
Why Neumann?	You don't CONTROL it; it's FREE to move
Force applied	Zero (natural boundary)

8. THE EQUATIONS (Weak Form)

Why Weak Form?

The strong form (differential equation) is hard to solve on a computer. The weak form converts it to integrals, which FEM can handle.

Thermal Equation (Weak Form):

Find T such that: $a_{\text{thermal}}(T, v) = L_{\text{thermal}}(v)$ for all test functions v

Left Side (a_{thermal}) - Terms with Unknown T :

• Transient (mass): $\rho \cdot c_p / dt \times T \times v \times \text{thickness} \times dx$ • Conduction: $k \times \nabla T \cdot \nabla v \times \text{thickness} \times dx$ • Convection: $h \times T \times v \times \text{thickness} \times ds(\text{edges})$

Right Side (L_thermal) - Known Terms:

• Old temperature: $\rho \cdot c_p / dt \times T_{old} \times v \times \text{thickness} \times dx$ • Ambient cooling: $h \times T_{ambient} \times v \times \text{thickness} \times ds(\text{edges})$ • Candle heat: $q_{candle} \times \text{thickness} \times v \times ds(5)$

Mechanical Equation (Weak Form):

Find u such that: $a_{\text{mechanical}}(u, v) = L_{\text{mechanical}}(v)$ for all test functions v

Left Side: Elastic stiffness (resistance to deformation)

$$\int \sigma_{\text{elastic}}(u) : \epsilon(v) \, dx$$

Right Side: Thermal stress (wants to expand)

$$\int \sigma_{\text{thermal}}(T) : \epsilon(v) \, dx$$

Key Idea: $\sigma_{\text{elastic}} = \sigma_{\text{thermal}}$ (elastic stress balances thermal stress)

9. WHAT IS ufl? (Unified Form Language)

ufl is a Python library that lets you write math equations as code.

ufl Function	What It Does	Example	Means
ufl.grad(T)	Compute gradient	∇T	$(dT/dx, dT/dy)$
ufl.dot(a, b)	Dot product	$a \cdot b$	$a_x \times b_x + a_y \times b_y$
ufl.tr(tensor)	Trace	$\text{tr}(\epsilon)$	$\epsilon_{xx} + \epsilon_{yy}$
ufl.inner(a, b)	Inner product	$a:b$	For tensors/matrices
dx	Volume integral	$\int \dots dx$	Sum over entire domain
ds	Surface integral	$\int \dots ds$	Sum over boundaries
ds(5)	Boundary 5 only	$\int \dots ds(5)$	Sum over boundary tagged 5

Why Use ufl?

- ✓ Write math naturally (looks like equations)
- ✓ Avoid manual gradient/derivative calculations (error-prone)
- ✓ DOLFINx converts it to FEM matrices automatically

Example:

```
k * ufl.dot(ufl.grad(T_trial), ufl.grad(v_T)) * thickness * dx
```

Translates to:

$$\int k \cdot (\nabla T \cdot \nabla v) \cdot \text{thickness} \, dx$$

Means: Heat conducted through the plate (thermal conduction term)

10. TIME STEPPING - THE SOLUTION LOOP

Your simulation runs in 600 time steps (600 seconds, 1 second each).

Step-by-Step Process:

FOR each time step from 0 to 600:

1. SOLVE THERMAL PROBLEM: • Solve: $a_{\text{thermal}}(T, v) = L_{\text{thermal}}(v)$ • Get: Temperature everywhere
2. CALCULATE THERMAL STRESS: • $\delta T = T - T_{\text{initial}}$ • $\epsilon_{\text{thermal}} = \alpha \cdot \delta T \cdot I$ (how much wants to expand) • $\sigma_{\text{thermal}} = 2\mu \cdot \epsilon_{\text{thermal}} + \lambda \cdot \text{tr}(\epsilon_{\text{thermal}}) \cdot I$
3. SOLVE MECHANICAL PROBLEM: • Solve: $a_{\text{mechanical}}(u, v) = L_{\text{mechanical}}(v)$ • Get: Displacement everywhere
- 4.

UPDATE FOR NEXT STEP: • $T_{old} = T$ (save current temperature) • Print: T_{min} , T_{max} , max displacement 5. LOOP to step +1

Why this order? Temperature drives deformation (thermal $\rightarrow$ mechanical), not the other way around.

11. STRESS & STRAIN - KEY CONCEPTS

Concept	Symbol	Definition	Physical Meaning
Strain	ϵ	Change in shape/size	How much is deformed
Stress	σ	Internal force per area	How hard it's resisting
Thermal Strain	$\epsilon_{thermal}$	$\alpha \cdot \Delta T$	Wants to expand by this much
Elastic Strain	$\epsilon_{elastic}$	σ/E	How much it actually deforms
Elastic Stress	$\sigma_{elastic}$	$E \cdot \epsilon_{elastic}$	How much it pushes back

The Balance:

1. Material heats up ($\Delta T > 0$) 2. Wants to expand: $\epsilon_{thermal} = \alpha \cdot \Delta T$ 3. But clamped on left side (can't expand there) 4. Internal stress builds: $\sigma_{thermal}$ 5. Material deforms to relieve stress: $\epsilon_{elastic}$ 6. Creates elastic stress: $\sigma_{elastic}$ 7. At equilibrium: $\sigma_{elastic} = \sigma_{thermal}$

Example Numbers:

Step	Value	Unit	Meaning
Temperature increase	476	$^{\circ}\text{C}$	From 24°C to 500°C
Thermal strain	0.008	—	Wants to expand 0.8%
Elastic stress from this	$\sim 1.3 \times 10^8$	Pa	Internal pressure builds
Max displacement	2.12	mm	Plate bends upward

12. QUICK REFERENCE CHECKLIST

Before Starting Your Simulation, Ask:

☐ What is being HEATED? (geometry) ☐ What material properties do I need? ☐ Where is heat APPLIED? (Neumann location) ☐ Where is heat LOST? (convection) ☐ What is CLAMPED? (Dirichlet location) ☐ What is FREE to move? (Neumann) ☐ Have I applied enough boundary conditions? ☐ Are any boundary conditions CONFLICTING?

Common Mistakes to Avoid:

- ✗ Mixing Dirichlet and Neumann at same boundary
- ✗ Over-constraining (too many fixed values)
- ✗ Under-constraining (no fixed points, solution not unique)
- ✗ Confusing flux with temperature
- ✗ Forgetting the factor of 2 for top/bottom surfaces

Green Flags (You Did It Right!):

- ✓ Thermal BC: All Neumann (temperature is free to vary)
- ✓ Mechanical BC: Has at least 1 Dirichlet (fixed point)
- ✓ Heat source and cooling are on different boundaries

✓ Solution converges and makes physical sense

13. COMPLETE SUMMARY TABLE

Boundary	Location	Type	Equation/Code	Physical Meaning
Candle	Left edge, 20mm center	Neumann	$L += q_candle \times t \times v \times ds(5)$	20,000 W/m ² heat input
Convection (edges)	Left (no candle), right, top, bottom	Robin/Neumann	$a += h \times T \times v \times t \times ds(1,2,3,4)$ $L += h \times T_amb \times v \times t$	Cool to 24°C
Convection (surfaces)	Top & bottom faces	Robin/Neumann	$a += 2h \times T \times v \times dx$ $L += 2h \times T_amb \times v \times dx$	2× because 2 surfaces
Clamped (mech)	Left edge	Dirichlet	$bc = \text{dirichletbc}(0, \text{left})$	Position fixed, $u=0$
Free (mech)	Right, top, bottom	Neumann (implicit)	No BC	Can move freely

14. FINAL TIPS FOR RUNNING SIMULATION

Tip 1: Think Physically First

Ask 'What's ACTUALLY happening?' before thinking about equations. The physics guides your boundary conditions.

Tip 2: Separate Dirichlet from Neumann

At each boundary, make a binary choice: Do I CONTROL the value (Dirichlet) or the rate (Neumann)?

Tip 3: Check Your Mesh Quality

If results look wrong, it might be the mesh. Try increasing n_x and n_y (more elements = better resolution).

Tip 4: Verify Units

Always check: Heat flux [W/m²], Temperature [°C], Force [N], Displacement [m].

Tip 5: Start Simple, Build Up

Start with constant material properties, then add complexity (temperature-dependent k , radiation, etc.).